

Problem

Given that $F(s) = e^{-2s}(s + 1)$, then $f(t)$ is

- (a) $e^{-2(t-1)}u(t-1)$
- (b) $e^{-(t-2)}u(t-2)$
- (c) $e^{-t}u(t-2)$
- (d) $e^{-t}u(t+1)$
- (e) $e^{-(t-2)}u(t)$

Step-by-step solution

Step 1 of 1

Given $F(s) = \frac{e^{-2s}}{s+1}$

Therefore from time shift formula

$$f(t) = L^{-1}[e^{-as}F(s)] = f(t-a)u(t-a)$$

Hence here $a = 2$ and $F(s) = \frac{1}{s+1}$

$$f(t) = e^{-t}$$

Therefore $f(t-a) = e^{-(t-a)}$

$$\boxed{f(t) = e^{-(t-2)}u(t-2)}$$